Hayatullah Ahmadzai

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Website | Google Scholar

Education

2015–2019: Ph.D. in Economics (focus: Agricultural and Development Economics)
University of Nottingham, Nottingham, UK

2008–2011: M.S. in Agricultural Economics
Purdue University, West Lafayette, IN, USA

2002–2005: B.S. in Agronomy and Soil Sciences
Kabul University, Kabul, Afghanistan

Primary Research Interests: Agricultural & Food Policy • Climate Shocks, Adaptation, & Agricultural Resilience • Farm & Household Economics • Food Security and Rural Livelihoods • Applied Econometrics & Causal Inference

Academic and Research Appointments

06/2022–06/2025: Research Fellow
Centre for Research in Economic Development and International Trade (CREDIT), School of Economics, University of Nottingham, Nottingham, UK.

04/2019–04/2022: Postdoctoral Research Fellow
International Center for Biosaline Agriculture, Academic City, Dubai, UAE.

02/2012–09/2015: Senior Economist/Head of Monitoring, Evaluation & Learning (MEL)
Department for International Development (DfID, now FCDO), Kabul, Afghanistan.

03/2011–09/2015: Assistant Professor (Teaching-focused)
Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan.
(Retained faculty appointment alongside DfID role, 2012–2015; 1–2 courses per term.)

07/2011–02/2012: Economist and Value-Chain Development Specialist
FLAG International Afghanistan (FIA), Kabul, Afghanistan.

03/2006–10/2007: Teaching Assistant
Department of Agronomy and Soil Sciences, Kabul University, Kabul, Afghanistan.

Research Affiliations and Fellowships

2025–2027: MESA Global Academy Scholar
Middle East Studies Association (MESA), Washington, DC, USA (remote; based in Durham, NC, USA).

Competitive two-year fellowship (2025–2027) supporting policy-relevant research on Middle East and North Africa issues and engaging with regional researchers through seminars, workshops, and talks.

06/2025–Present: External Research Fellow (remote, unpaid)
Centre for Research in Economic Development and International Trade (CREDIT), University of Nottingham, Nottingham, UK (remote; based in Durham, NC, USA).

Continuing collaborative research on gender, conflict, and rural household welfare in Afghanistan, including a working paper on security, infrastructure, and the gender gap in rural labor markets using nationally-representative repeated cross-sectional survey data.

Publications

Peer-reviewed Journal Articles

Tests for market failures and overcoming transaction costs barriers to market participation in Afghanistan
Ahmadzai, H. 2026. *Journal of Social and Economic Development*, 28, 637–655.

Climate shocks, household food security and welfare in Afghanistan
Ahmadzai, H., and Morrissey, O. 2025. *Food Policy*, 134, 102910.

Crop Diversification, Household Welfare and Conflict: Afghanistan 2011–2017
Ahmadzai, H., and Morrissey, O. 2025. *The Journal of Development Studies*, 61(3), 381–399.

Empirical Analysis of Impacts of Sand and Dust Storms on Agricultural Productivity in Mongolia
Ahmadzai, H., Malhotra, A., and Tutundjian, S. 2023. *PLOS ONE*, 18(2), e0269271.

The Impact of Sand and Dust Storms on Agriculture Productivity in Iraq
Ahmadzai, H. 2023. *Middle East Development Journal*, 15(1), 50–65.

Hope for Change: Is Diversifying Production Portfolios an Ideal Strategy to Boost Farming Efficiency in Afghanistan?
Ahmadzai, H. 2022. *Progress in Development Studies*, 22(1), 7-31.

Orphan Crops: A Best Fit for Dietary Enrichment and Diversification in Highly Deteriorated Marginal Environments
Talabi, A., Vikram, P., Thushar, S., Hifzurrahman, R., Ahmadzai, H., Nhamo, N., Shahid, M., and Singh, R. 2022. *Frontiers in Plant Science*, 13, 839704.

Policies for Sustainable Agriculture and Livelihood in Marginal Lands: A Review
Ahmadzai, H., Elouafi, I., and Tutundjian, S. 2021. *Sustainability*, 13(16), 8692.

Development of Quinoa Value Chain to Improve Food and Nutritional Security in Rural Communities in Rehamna, Morocco: Lessons Learned and Perspectives
Hirich, A., Rafik, S., Rahmani, M., Fetouab, A., Filali, K., Ahmadzai, H., et al. 2021. *Plants*, 10(2), 301.

How is Off-farm Income Linked to On-farm Diversification? Evidence from Afghanistan
Ahmadzai, H. 2020. *Studies in Agricultural Economics*, 122(1), 1–12.

Trends in Quinoa Adoption in Marginal Areas, Economic Viability, and Policy Outlook
Ahmadzai, H. 2020. *Journal of Agribusiness and Rural Development*, 57(3), 235–247.

Book Chapters

Marginal Areas and Indigenous People Priorities for Research and Action
Sayed, A., and Ahmadzai, H., et al. 2023. In *Science and Innovations for Food Systems Transformation*, edited by von Braun, J., et al.,
pp. 261–279. Springer International Publishing.

Impacts of Sand and Dust Storms on Agricultural Production
Middleton, N., Ahmadzai, H., and Ayurzana, E. 2023. In *Sand and Dust Storms: A Guide to Mitigation, Adaptation, Policy and Risk Management Measures in Agriculture*, edited by Middleton, N., pp. 33–49. FAO.

From Desert Farm to Fork: Value Chain Development for Innovative Salicornia-based Food Products in the United Arab Emirates
Lyra, D., Lampakis, E., …, Ahmadzai, H., et al. 2021. In *Future of Sustainable Agriculture in Saline Environments*, edited by Negacz, K., et al.,
pp. 181–199. CRC Press/Taylor & Francis Group.

Policy Papers & Technical Reports

Marginal Lands: Potential for Agricultural Development, Food Security and Poverty Reduction
Ahmadzai, H., Tutundjian, S., et al. 2022. *SOLAW21 Technical Background Report*, Rome: FAO.

Agri-tech: A Catalyst for Treated Wastewater Deployment in High Efficiency Food Production Systems
Ammar, K., Fatnassi, H., and Ahmadzai, H., et al. 2022. *International Center for Biosaline Agriculture (ICBA)*, Dubai, UAE.

Marginal Areas and Indigenous People: Priorities for Research and Action
Sayed, A., and Ahmadzai, H., et al. 2022. *Food Systems Summit Brief.* Food Systems Summit 2021.

Addressing Sand and Dust Storms in Northeast Asia (China, Mongolia), Western Asia (Iran, Iraq, and Kuwait), and North Africa (Algeria): An Agricultural Perspective
Ahmadzai, H. 2021. *Food and Agriculture Organization (FAO) and International Center for Biosaline Agriculture (ICBA).*

Baseline Survey Report: Improving Agricultural Resilience to Salinity (RESADE)
Ahmadzai, H. 2020. International Center for Biosaline Agriculture (ICBA), IFAD, BADEA.

Working Papers

Security, Infrastructure, and Gender Gap in Rural Labor Markets: Evidence from Afghanistan's Repeated Cross-Sectional Surveys
Ahmadzai, H. 2026.
CREDIT Research Paper; Work in Progress

Research Projects and Funding

2022–2024: Institute of International Education (IIE), CARA (Council for At-Risk Academics), and the University of Nottingham, IIE-SRF Afghanistan Emergency Initiative Fund. **Principal Investigator**. Research Support: £86,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Catalyzing Investments and Actions to Enhance Resilience Against Sand and Dust Storms in Agriculture. **Principal Investigator**. Grant Amount: $150,000.

2021: Food and Agriculture Organization of the United Nations (FAO), Climate-Resilient and Gender-Inclusive Cropping Systems in Climate-Vulnerable Areas in Asia and the Pacific. **Co-Principal Investigator** with R.K. Singh (ICBA). Total Project Budget: $178,441.

2021: Ministry of Climate Change and Environment, UAE, Developing National Policy to Utilize Treated Sewage Effluent (TSE) and Biosolids in Controlled Environment Agriculture in the UAE. **Co-Principal Investigator** with Khalil Ammar (ICBA). Total Project Budget: $100,000.

2020: International Center for Biosaline Agriculture (ICBA), Adoption and Impact of Quinoa Among Farmers in Marginal Areas of the MENA Region. **Principal Investigator**. Grant Amount: $68,488.

2020: University of Wollongong in Dubai and ICBA, Resilient Agrifood Dynamism Through Evidence-Based Policies (READY). **Co-Principal Investigator** with Ioannis Manikas (University of Wollongong) and Khalil Ammar (ICBA). Total Project Budget: $1,500,000.

2019: IFAD and BADEA, Improving Agricultural Resilience to Salinity and Climate Change through Development of Pro-Poor Technologies and Management Strategies (RESADE). **Co-Principal Investigator** with R.K. Singh (ICBA). Total Project Budget: $7,700,000.

2019: International Center for Biosaline Agriculture (ICBA), EXPO Live Phase II: Value Chain Development for Innovative Halophyte-Based Food Systems. **Co-Principal Investigator** with Dionysia Angeliki Lyra (ICBA). Total Project Budget: $500,000.

Teaching Experience

2011–2015: Assistant Professor (teaching-focused), Department of Agricultural Economics and Extension, Kabul University, Kabul, Afghanistan (full-time; concurrent with DfID role, 2012–2015).

Teaching responsibilities:

Agricultural Production Economics (AgEcEx731), undergraduate; spring terms, 2011–2015.

Agricultural Finance (AgEcEx836), undergraduate; fall terms, 2011–2015.

Introductory Statistics & Econometrics(AgEcEx630), undergraduate; spring terms, 2012–2014.

Research Methods & Projects (AgEcEx735; seminar-based teaching and student research supervision), undergraduate; spring & fall terms, 2011–2015.

Selected Presentations & Talks

Invited Talks

2026: "Climate Shocks, Household Food Security, and Welfare in Afghanistan." Stanford University (FSE, Center for South Asia, Abbasi Program in Islamic Studies), April 22, 2026. Event | Slides | Video

2021: "Challenge and Opportunity of Quinoa for Food and Nutrition Security." Panellist with Warren T.K. Lee, chaired by Bo Zhou. FAO Regional Office for Asia and the Pacific (FAO-RAP), Bangkok, Thailand, November 7–8.

2021: "The Economics of Quinoa Production: Opportunity for Marginal Farmers." Panellist with Dr. José Graziano da Silva. Global Forum on Innovations for Marginal Environments (GFIME), Dubai, UAE, November 20–21.

Conference Presentations

2023: "Crop Diversification and Farm Household Welfare during Conflict: Afghanistan 2011–2017." 6th MaDE Workshop on Emerging and Developing Countries, Loughborough Business School (online), June 5–6, 2023. Slides

2021: "Assessing the Impact of Sand and Dust Storms on Agriculture." Food and Agriculture Organization (FAO), Bangkok (online), April 7–11, 2021. Slides

2020: "Salinization in Marginal Agricultural Areas: Implications for Global Food Security and Poverty." International Center for Biosaline Agriculture (ICBA), Dubai, UAE, June 27–28, 2020. Slides

2018: "Tests for Market Failures and Overcoming Transaction Costs in Afghanistan." Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, September 8–12, 2018. Slides

2017: "Status, Patterns, and Microeconomic Drivers of the Extent of Diversity in Crop Production: Evidence from Afghanistan." Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 15–20, 2017. Slides

2016: "Crop Diversification and Technical Efficiency: Evidence from Afghanistan." Ph.D. Conference, School of Economics, University of Nottingham, Nottingham, UK, November 12–16, 2016. Slides

Awards

2021: Co-recipient, Khalifa International Award for Date Palm and Agricultural Innovation (KIADPAI), awarded as Co-PI of the Project: "From Desert Farm to Fork: Value Chain Development for Innovative Halophyte-Based Food Products."

2015: PhD Scholarship Award, University of Nottingham and The Islamic Development Bank (IsDB).

2007: Graduate Scholarship Award, Purdue University and USAID-funded Afghan Merit Scholar Program (AMSP).

Service and Professional Activities

2019–present: Member, Agribusiness Sub-network, Association of Agricultural Research Institutions in the Near East and North Africa (AARINENA).

2019–present: Peer Reviewer for *The Journal of Development Studies*; *Food Policy*; *Sustainability* (MDPI); *PLOS ONE*; *Review of Development Economics*; *Humanities and Social Sciences Communications*; *Agriculture* (MDPI); *Cogent Economics and Finance*; *Journal of Social and Economic Development.*

Media and Outreach

2022: "How COVID-19 Crisis Undermined MENA States' Food Security." *Arab News*, November 20, 2022.

2021: "Agriculture in Marginal Lands." *Scholarly Community Encyclopedia*, September 2021.

2021: "New Study Reviews Status of Marginal Lands Worldwide." ICBA News Article, August 29, 2021.

2021: "World's Indigenous Peoples to Bear Brunt of Climate Change, Warns UN Paper." ICBA News Article, April 25, 2021.

2020: "ICBA's Quinoa Varieties Show Promise in Morocco." ICBA News Article, April 15, 2020.

Professional Development

2020: Responsible Conduct in Research: Research, Ethics, and Compliance. Collaborative Institutional Training Initiative (CITI). Completed online, March 24–April 4.

2013: Leadership and Governance of Development Programs and Projects. Crown Agents International. Dubai, United Arab Emirates, November 11–15.

2012: Management of Agriculture and Rural Development Projects. Asian Institute of Technology (AIT). Bangkok, Thailand, November 21–December 1.

2008: Academic English and Cross-Cultural Training Program. International Centre for Intercultural Communication (ICIC). IUPUI, Indianapolis, IN, USA, October 29, 2007–May 30, 2008.

Research Methods and Skills

- **Econometrics & Causal Inference:** Applied econometrics; causal inference; quasi-experimental methods (IV, DiD, RDD, matching); panel-data methods; discrete-choice models; experimental and field-experimental methods; welfare and distributional analysis.
- **Agricultural & Field Research:** Farm and household surveys; agricultural production and farm economics; farmer adoption and technology uptake; agricultural productivity; value-chain analysis; rural livelihoods; food security; field-based data collection and research design.
- **Data & Empirical Research:** Survey sampling and design; administrative, climate, longitudinal, and geospatial data; dataset construction and harmonization; quantitative research design; mixed-methods research; reproducible data and analysis workflows.
- **Research Software & Data Tools:** Stata, R, Python (Pandas, NumPy, Statsmodels, Seaborn, Matplotlib), QGIS, Survey Solutions, Qualtrics, Excel.
- **Research Management:** Grant writing and proposal development; multidisciplinary research collaboration; project management; research supervision and mentoring; technical and scientific reporting.

Linguistic

- English (fluent)
- Pashto (native)
- Dari/Persian (fluent)
- Arabic (basic)